ALI AUN HAIDER

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PROFESSIONAL SUMMARY

Full-stack and AI engineer who ships production applications end to end: React/JavaScript front ends, Flask/Python REST APIs, MySQL data models, and LLM-powered features including RAG pipelines, agentic workflows, and evaluation harnesses. Backed by strong CS fundamentals and a Computer Science degree earned Magna Cum Laude. Two years auditing web accessibility trained me to read HTML, CSS, and ARIA at the source, debug to root cause, and hold quality to a measurable standard, the same rigor I bring to grounding, validating, and shipping AI systems.

PROJECTS

AI Tutor Phase 2: Full-Stack RAG Tutoring Platform *(Senior Capstone, Primary Developer)*
Tech: React, Flask/Python, MySQL, SQLAlchemy, OpenAI, CAS SSO, GitLab CI/CD, pytest, Docker

- Served as primary developer of a course-aware AI tutoring platform deployed to production for Miami University, delivered as a five-person senior capstone for real university clients: React front end, Flask REST API, and MySQL/SQLAlchemy data models across 10+ related tables.
- Built the AI layer as a teaching assistant rather than an answer engine: course-scoped content retrieval, per-request classification, academic-integrity guardrails, and structured response metadata that powers faculty analytics.
- Integrated CAS single sign-on with role-based access (student/faculty/admin), server-side sessions, and production safeguards (secret-key validation, Secure/HttpOnly cookies, environment-gated dev bypass); shipped through a GitLab CI/CD build-and-deploy pipeline with a pytest backend suite and a health endpoint.

AI Job Application Agent: Autonomous Agent Pipeline
Tech: Python, Playwright, deterministic scoring, LLM-grounded generation

- Designed an autonomous agent with a clean perception, reasoning, memory, execution, and orchestration architecture that ingests jobs, scores fit deterministically, and drives a controller loop with persistent, retry-aware state.
- Built Playwright browser automation with deterministic field mapping and dry-run autofill for Greenhouse and Lever style forms, plus grounded, per-job answer and resume tailoring that reorders and emphasizes real content without fabricating anything.
- Engineered a guarded-submission policy that keeps submission off by default and blocks on any unresolved sensitive field, with auditable review import/export and run-level tuning and diagnostics reports.

University Syllabus RAG Chatbot: Retrieval-Augmented Q&A
Tech: Python, ChromaDB, hybrid retrieval + reranking, OpenAI, Streamlit, Docker

- Built a source-grounded document Q&A app over syllabus and policy PDFs and DOCX, with ingestion, configurable chunking, embeddings, and a local ChromaDB vector store.
- Implemented hybrid dense-plus-keyword retrieval with optional reranking and citation-enforced answers, plus a retrieval-only mode that runs without model quota.
- Wrote a retrieval evaluation harness reporting Precision@K and Recall@K to measure and tune quality; delivered a Streamlit UI and containerized the app with Docker.

HOME NEXUS: Full-Stack Real Estate Web Application *(Team Lead, 4 engineers)*
Tech: Flask, Python, SQLAlchemy, HTML/CSS/JavaScript

- Led a four-person team building a Flask/SQLAlchemy property-listings platform with user authentication, multi-criteria and address-based search, and a favorites system with add/remove and profile management.
- Divided tasks, planned the work, and coordinated front-end and back-end development to delivery, building the interface with semantic, standards-based markup.

EXPERIENCE

Web Accessibility Tester (Intern) — **Apr 2024 to May 2026**
Miami University, IT Services — *Oxford, OH*

- Audited hundreds of digital platforms, websites, and content against WCAG 2.1 A/AA (Workday, Zoom, Slack, Webex, Handshake), combining manual testing with automated scanners across macOS, Windows, iOS, and Android.
- Read the underlying HTML, CSS, and ARIA to trace each defect to its root cause instead of stopping at scanner output, the same source-level debugging I apply to full-stack code and to grounding and validating AI outputs.
- Tested with screen readers (NVDA, JAWS), keyboard-only flows, and browser dev tools; wrote criterion-mapped defect reports pairing every finding with a concrete remediation, drove fixes to closure with IT under real security and compliance constraints, and trained and onboarded new hires on the methodology.

ADDITIONAL PROJECTS

Model Evaluation & Benchmarking Pipeline: automated pipeline benchmarking 7 model architectures on a shared dataset (~94% accuracy) with hyperparameter tuning, k-fold cross-validation, and F1/ROC-AUC scoring (Python, scikit-learn, XGBoost, pandas).
Classical Cell Segmentation: trainable pixel-level segmentation on Cell Tracking Challenge data with a reproducible train/validate pipeline (Python, scikit-learn, computer vision).
Streaming Services Explorer: catalog search and genre analytics over a relational store with prepared-statement queries (Java, MySQL, JDBC).

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash, HTML, CSS
AI / ML: LLM integration, RAG pipelines, embeddings & vector stores (ChromaDB, FAISS), agentic workflows, prompt engineering, output grounding & validation, evaluation (Precision@K/Recall@K, F1, ROC-AUC), scikit-learn, XGBoost, pandas, NumPy
Web & Backend: React, Node.js, Flask, REST APIs, SQLAlchemy, authentication & SSO (CAS), role-based access, MySQL / SQL Server / Oracle, relational database design
Tools & Practices: Git / GitHub / GitLab, Docker, CI/CD, Playwright, pytest, unit testing, Agile/Scrum, Linux; WCAG 2.1, ARIA, semantic HTML, NVDA/JAWS, Axe DevTools

EDUCATION

Miami University — Oxford, OH
Bachelor of Science in Computer Science | Minor: Mathematics — May 2026
GPA: 3.83, Magna Cum Laude | President's List and Dean's List
Relevant Coursework: Machine Learning, Generative AI, Artificial Intelligence, Web Application Programming, Database Systems, Data Structures & Algorithms, Software Engineering, High Performance Computing